Economic Fibrillation

Complete Figure Atlas

Every graph and diagram from the real-data analysis, in sequence — from turning GDP into oscillators, through validation, to the tested fibrillation signature around global recessions.

1 · Turning GDP into Oscillators — the Method

Every economy's GDP rises over time, so it is not itself an oscillator. This section shows how the business-cycle wobble is extracted and converted into a phase, so economies can be treated as coupled oscillators.

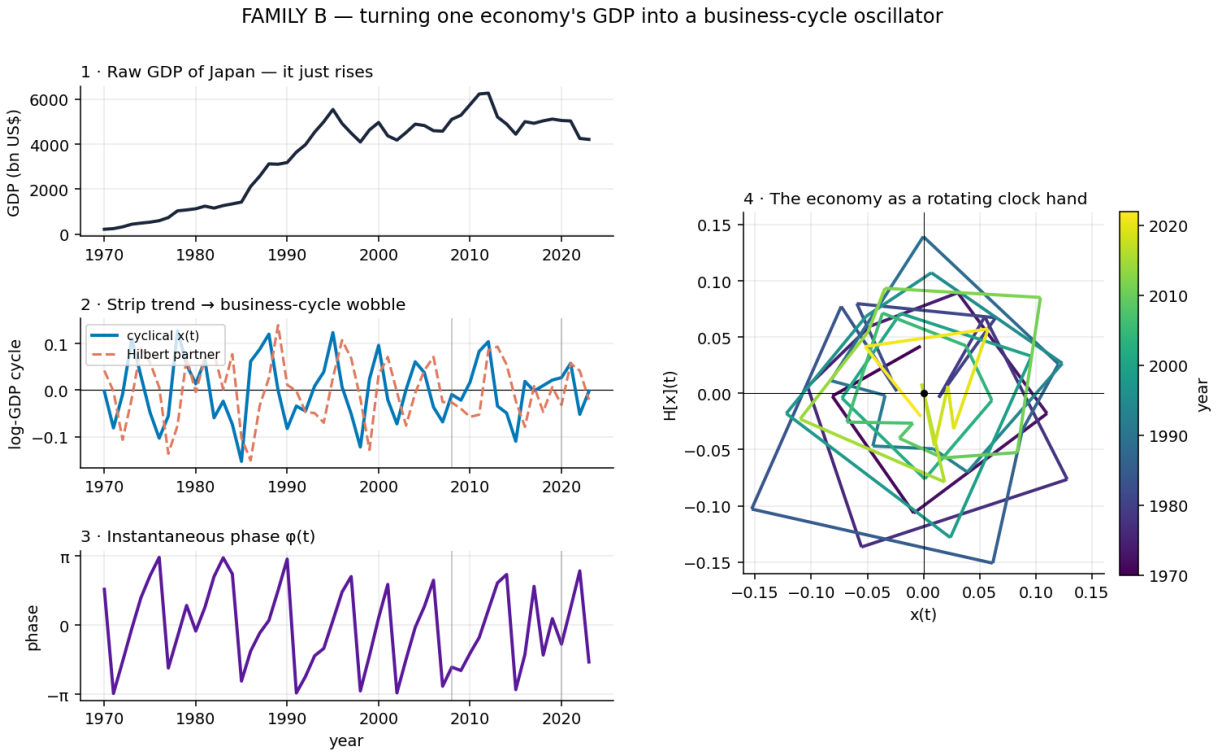


Fig 1. The Hilbert pipeline, one economy (Japan). Raw GDP rises smoothly (top); removing the trend leaves the business-cycle wobble and its 90°-shifted Hilbert partner (middle); the phase is the clock-hand angle (bottom); and the analytic signal traces a rotating phasor in the complex plane (right). **Implication:** a rising, non-oscillatory GDP series can be rigorously converted into a well-defined cyclical phase — the foundation for treating economies as coupled oscillators.

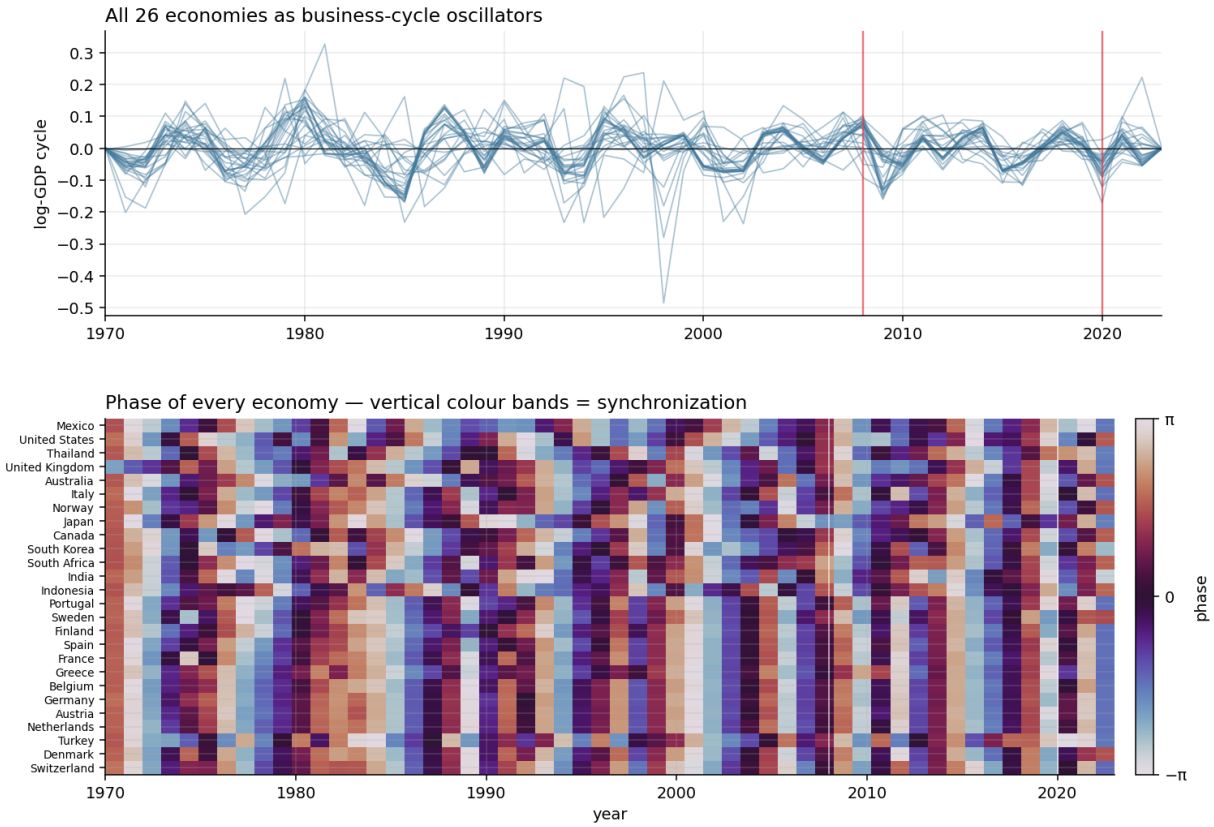


Fig 2. All 26 economies as oscillators. Top: every economy's cyclical component overlaid. Bottom: a phase heatmap, each row an economy, colour its cycle position. **Implication:** vertical bands of uniform colour mark years when economies share a phase — synchronization is visible directly in the data, and it concentrates around 2008 and 2020.

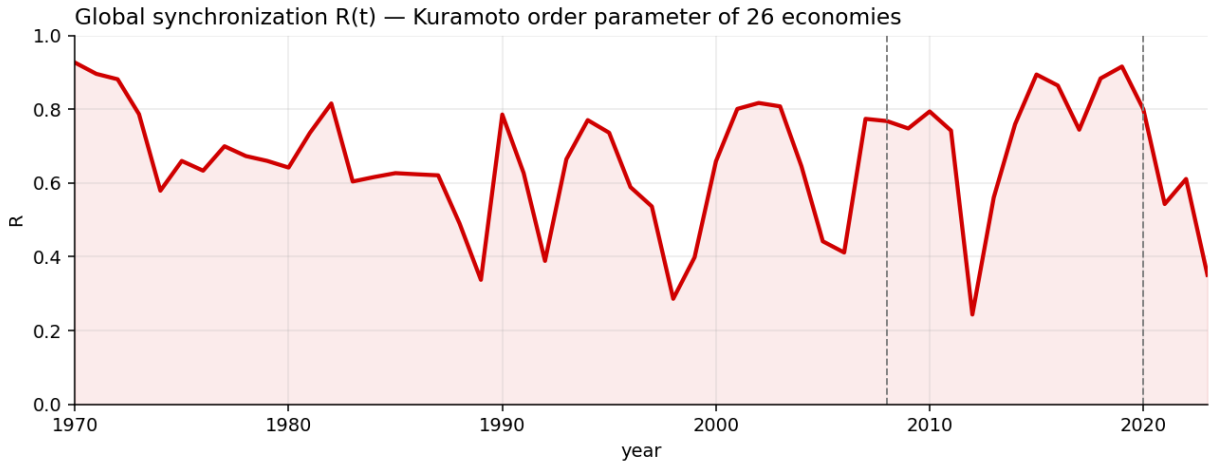


Fig 3. The Kuramoto order parameter R(t), 1970–2023. A single number summarising how aligned the 26 economies are each year (0 = scattered, 1 = locked). **Implication:** synchronization is not constant — it rises and falls over decades and spikes at global crises, motivating the entire study.

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2 · Validating the Signal — is it real?

Before interpreting synchronization, we test whether it exceeds what pure chance produces, using 5,000 Monte-Carlo surrogate 'null' worlds that preserve each economy's own rhythm but destroy all alignment.

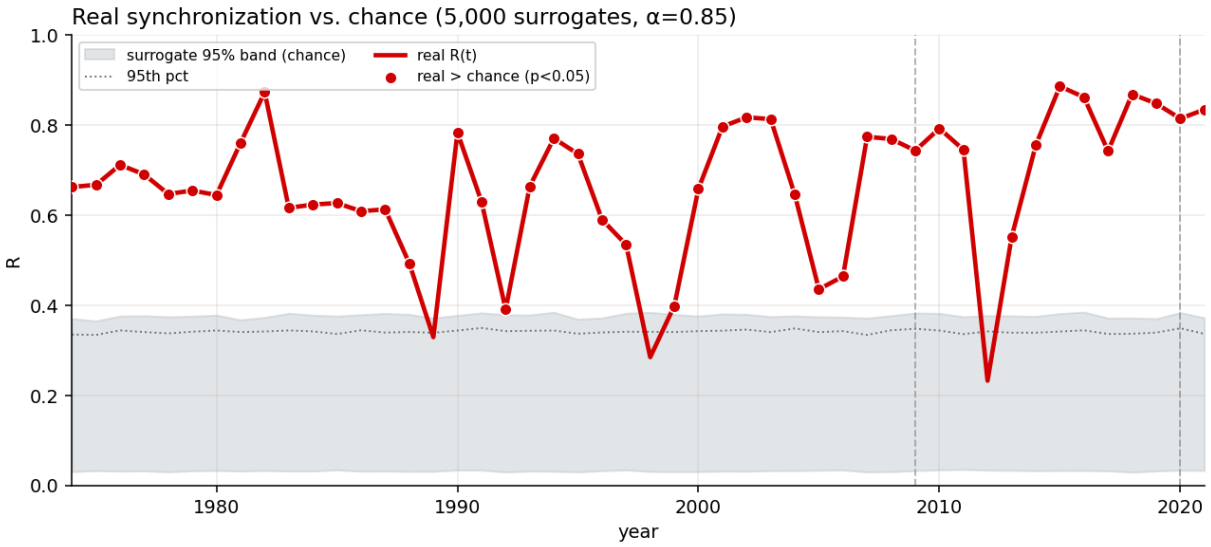


Fig 4. Real synchronization vs. chance. Red = observed R(t); grey band = the range from 5,000 surrogate worlds with alignment destroyed. **Implication:** real synchronization sits far above chance almost everywhere (p = 0.0002 at 2008/2020) — the signal is genuine, not a statistical accident.

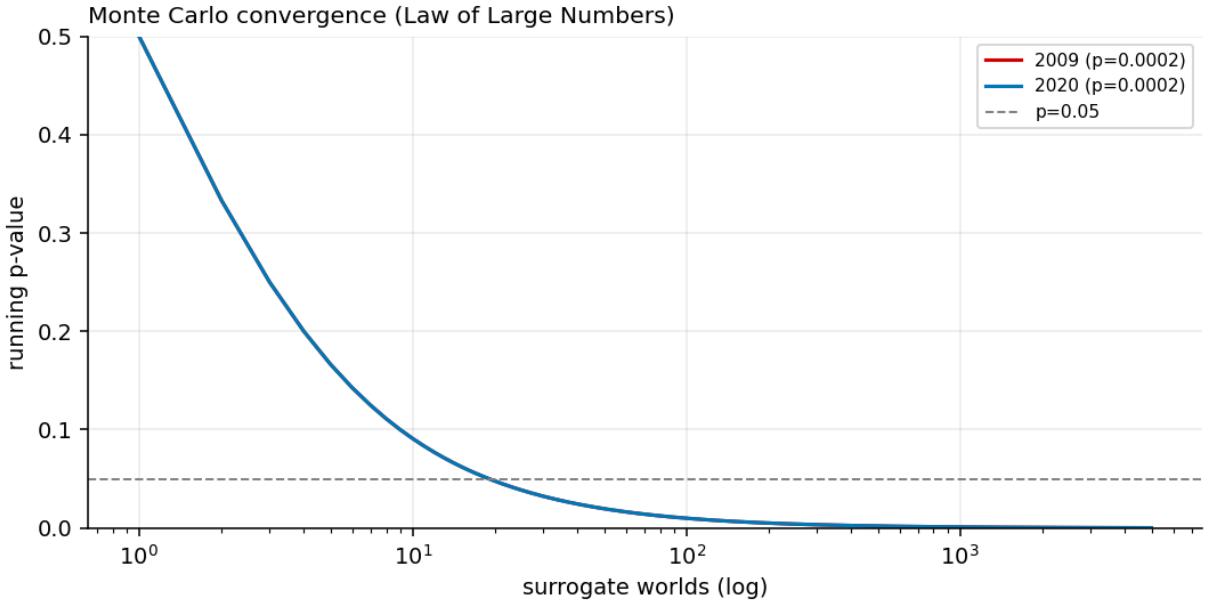


Fig 5. Monte-Carlo convergence. The estimated p-value for 2009 and 2020 as the surrogate ensemble grows. **Implication:** the p-value settles to a stable near-zero value — a direct demonstration of the Law of Large Numbers, confirming the ensemble is large enough.

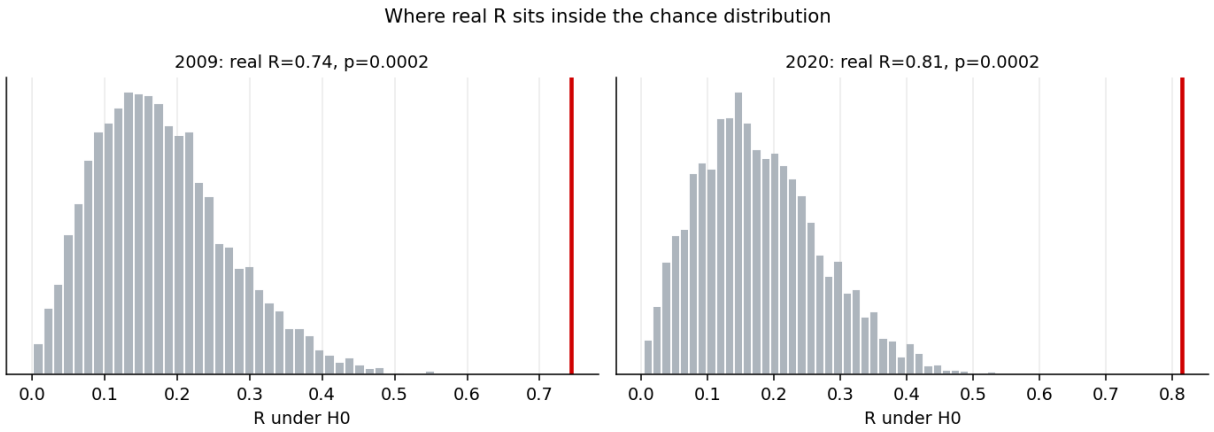


Fig 6. The null distributions. Histograms of synchronization under the no-coupling null for 2009 and 2020, with the real value (red) far in the tail. **Implication:** the observed crisis-year synchronization is essentially unreachable by chance.

3 · Crisis Analysis I — Current-Dollar GDP

A first pass comparing synchronization against crisis windows chosen from narrative history, using nominal (current US$) GDP.

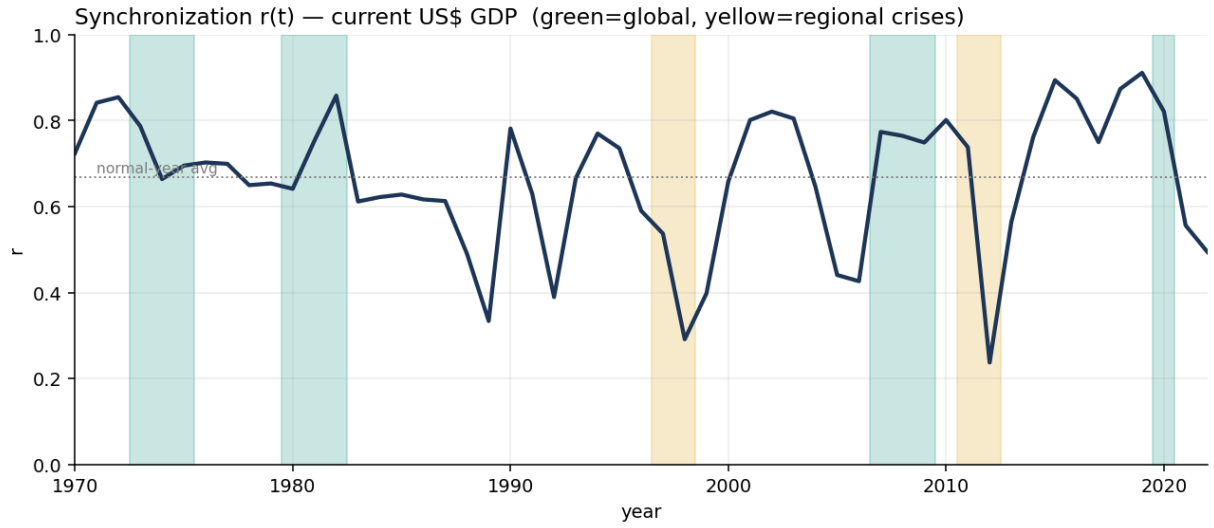


Fig 7. Synchronization with crisis windows (current-$ GDP). Green = four global crises, yellow = two regional crises. **Implication:** synchronization sits high in global crises and at troughs in regional ones — the first sign R(t) distinguishes the two.

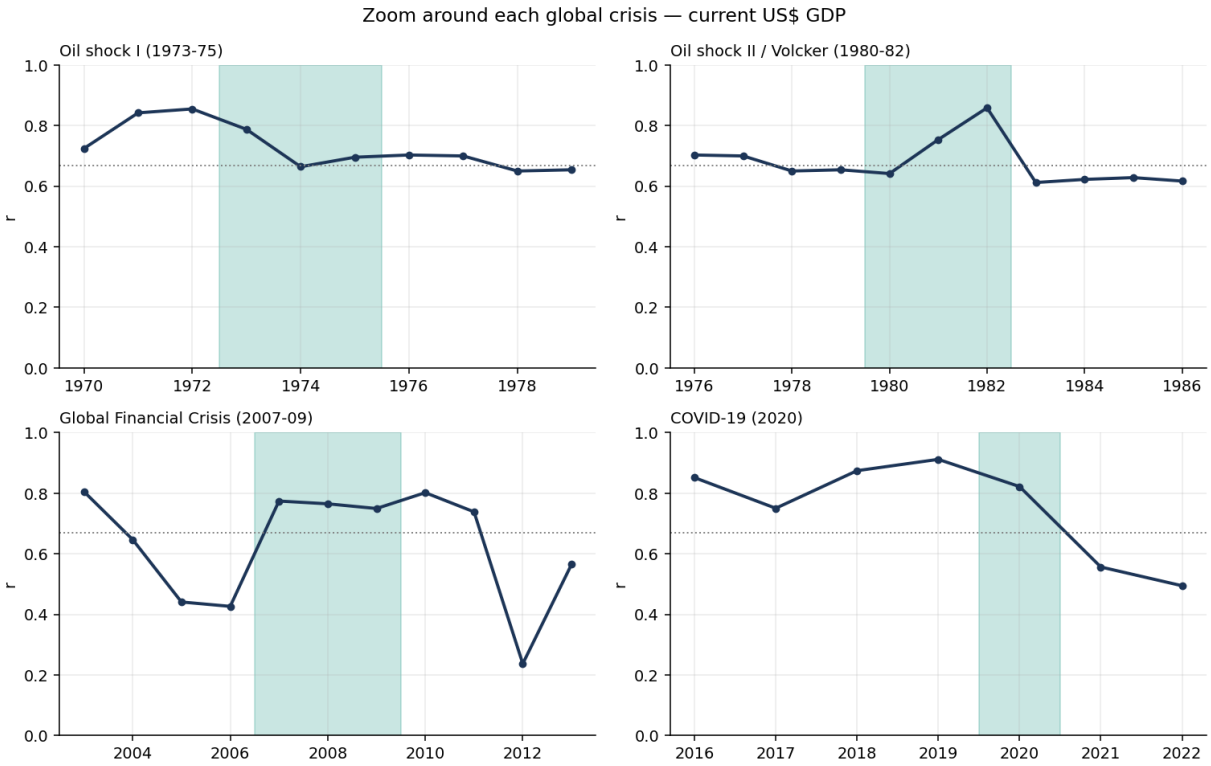


Fig 8. Each global crisis, zoomed. **Implication:** synchronization builds into the 1980–82 and 2007–09 crises and shatters after 2020 — the rise-then-rupture shape, crisis by crisis.

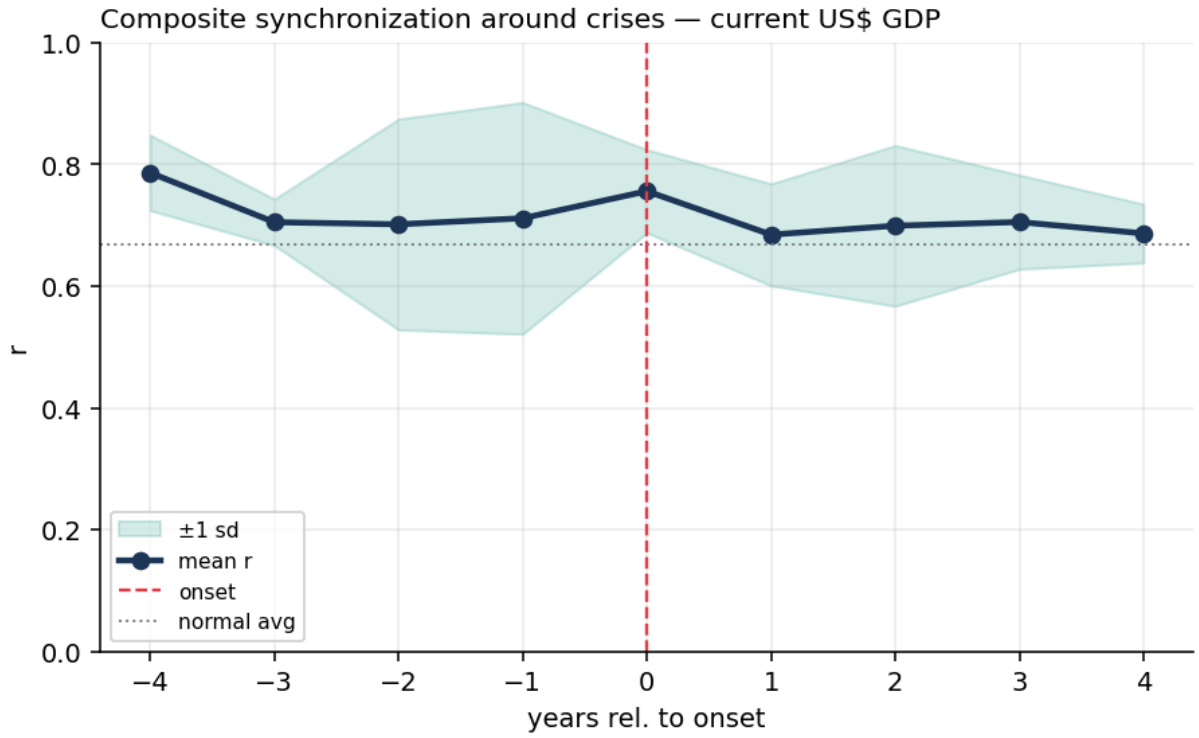


Fig 9. Composite synchronization around global-crisis onsets. **Implication:** on average, synchronization is elevated through the crisis window — the first quantitative hint of the fibrillation signature.

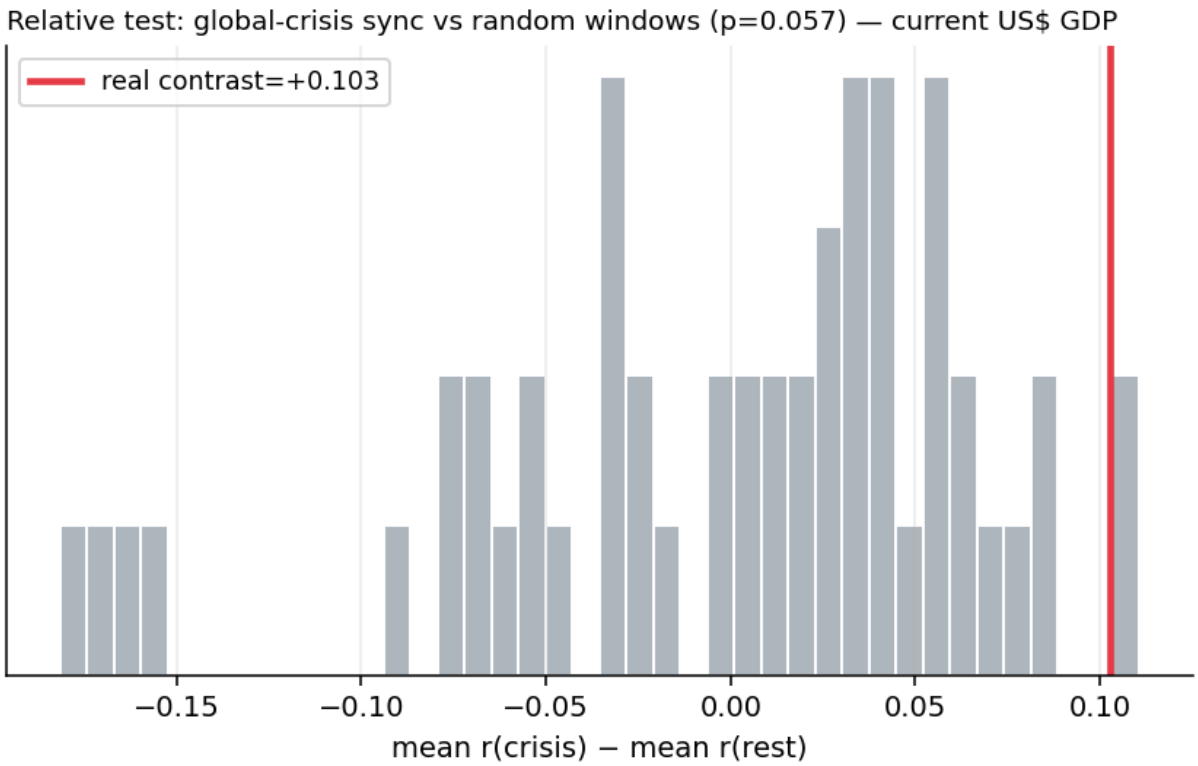


Fig 10. Relative test (current-$). Real global-crisis contrast vs. randomly placed windows. **Implication:** global crises are more synchronized than random placement gives, though only borderline (p ≈ 0.06) in nominal data.

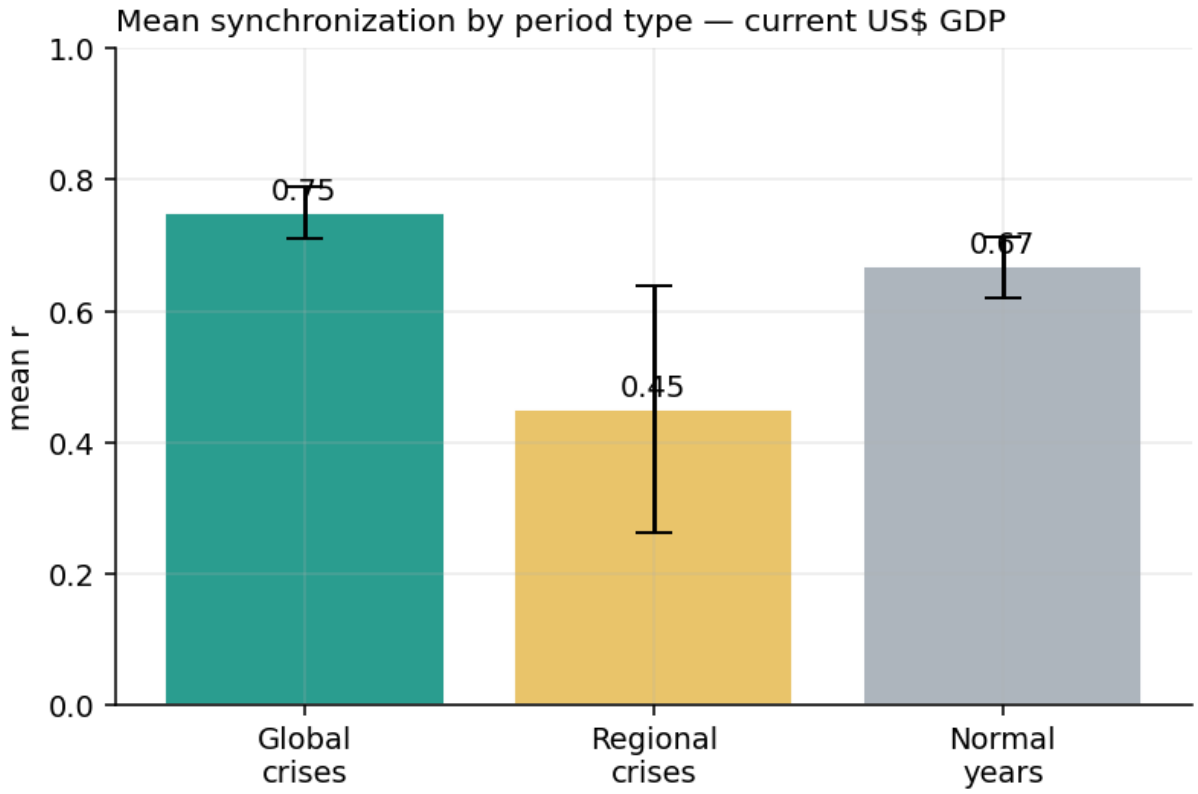


Fig 11. Mean synchronization by period type (current-$). **Implication:** global crises (0.75) > normal years (0.67) >> regional crises (0.45) — synchronization cleanly separates global from regional shocks.

4 · Crisis Analysis II — Real GDP (Robustness)

The same analysis repeated on real, inflation- and PPP-adjusted GDP (Maddison Project) to check the findings are not an artefact of inflation or exchange rates.

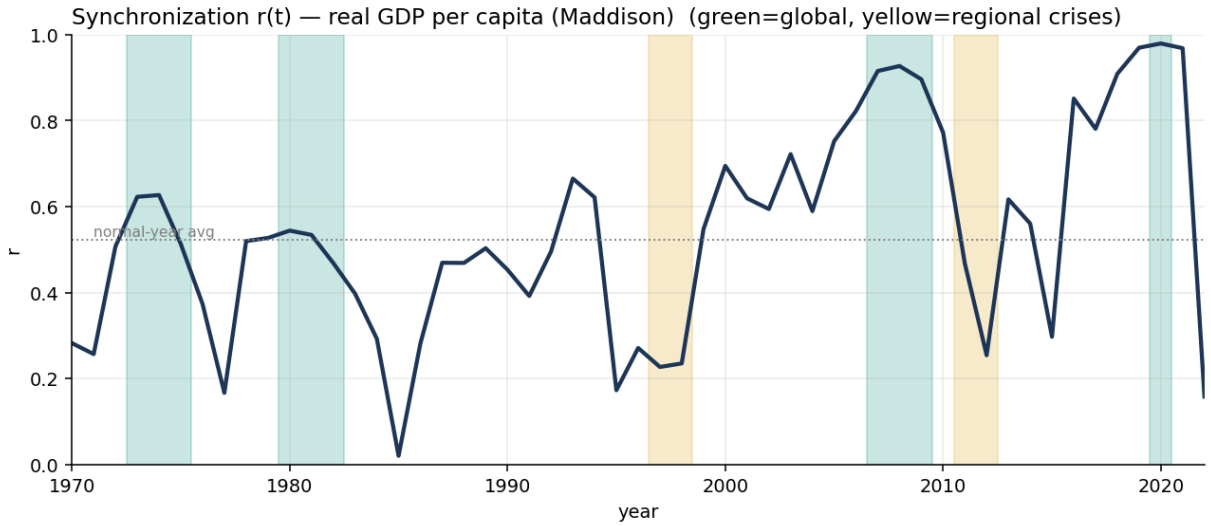


Fig 12. Synchronization with crisis windows (real GDP). **Implication:** constant-price data sharpens the picture — 2008 and 2020 stand out even more clearly as synchronization peaks.

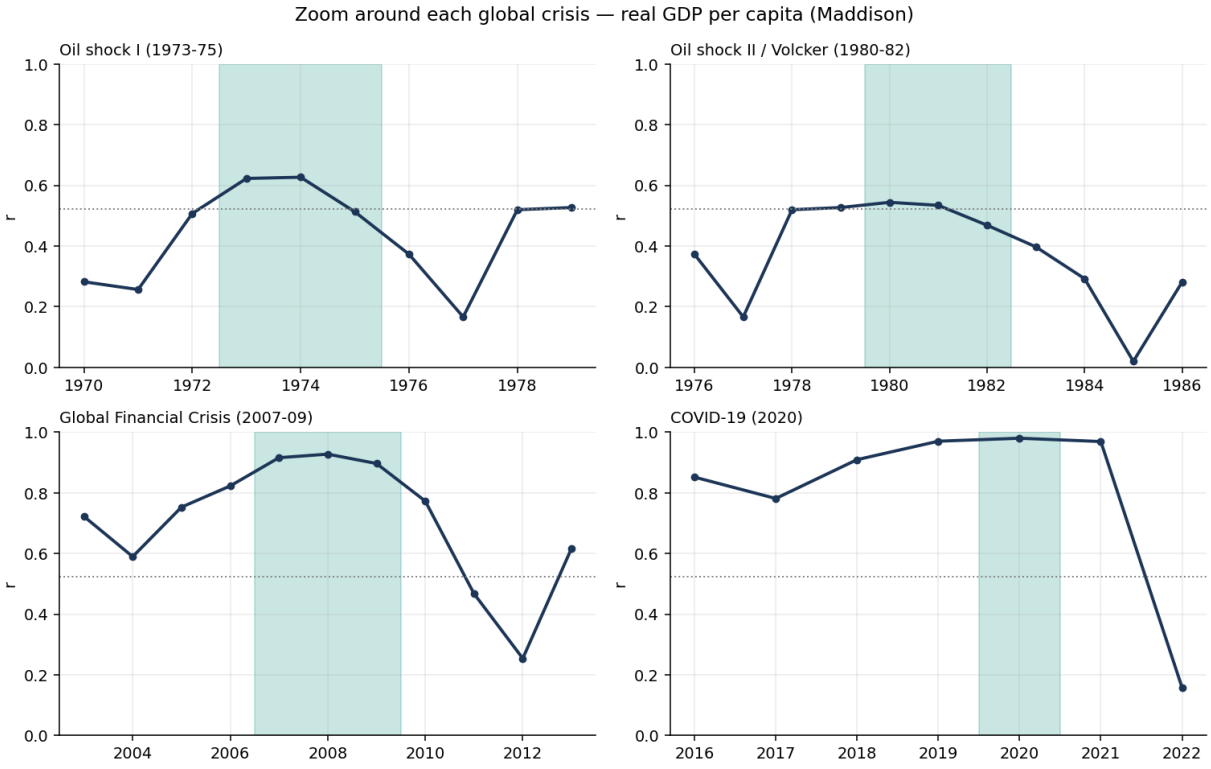


Fig 13. Each global crisis, real GDP. **Implication:** the 2007–09 and 2020 shapes are cleaner in real terms; the oil shocks are muted, revealing their nominal synchronization was partly inflation.

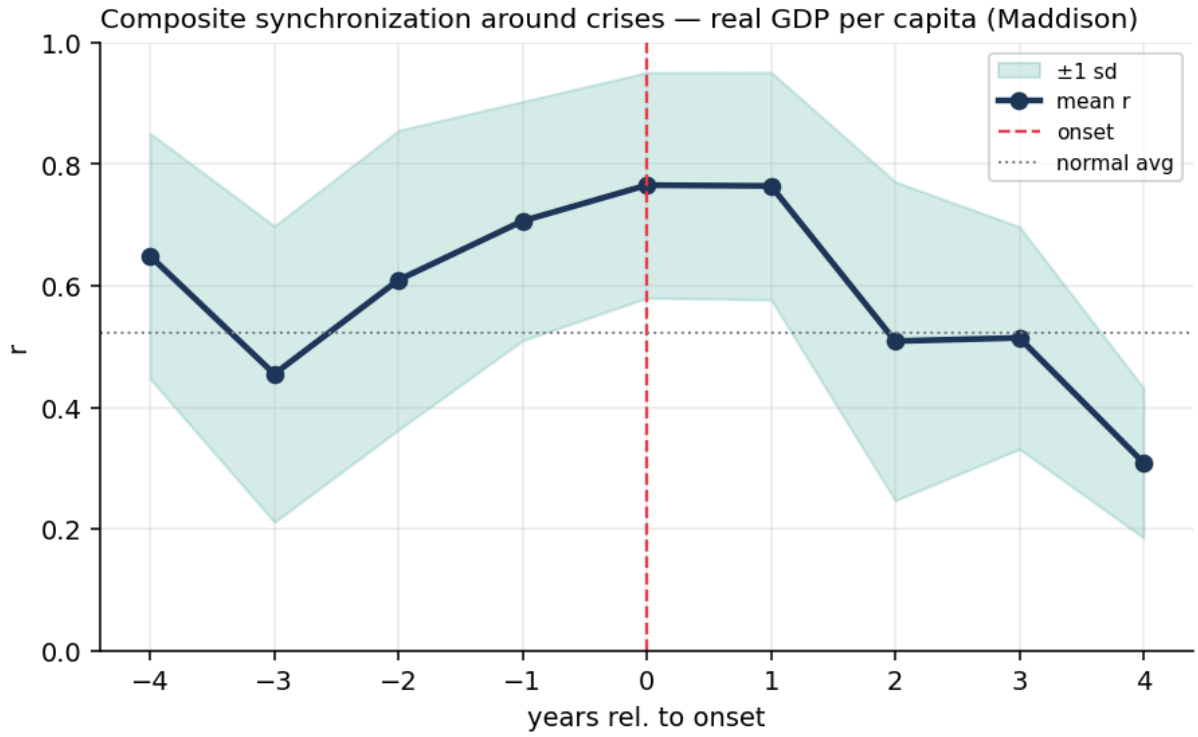


Fig 14. Composite, real GDP. **Implication:** the elevated-around-onset shape survives the more demanding data — the signature is not a nominal artefact.

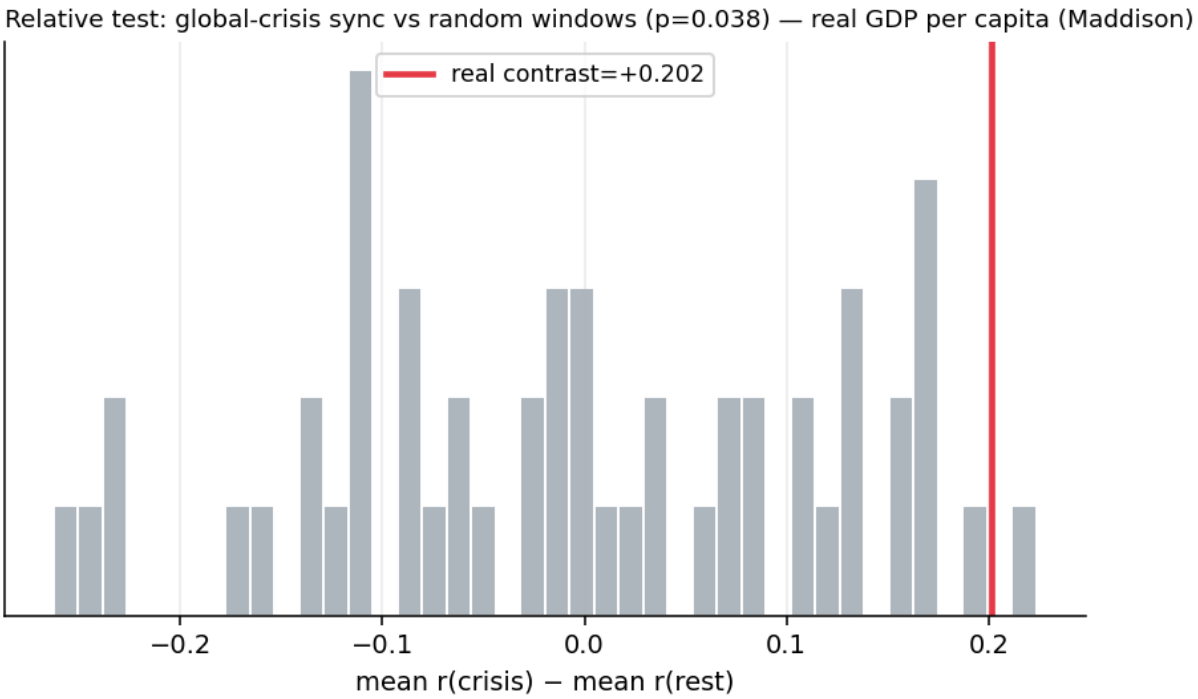


Fig 15. Relative test, real GDP. **Implication:** the global-crisis contrast is now significant (p = 0.038) and larger than in nominal data — the result strengthens under real GDP.

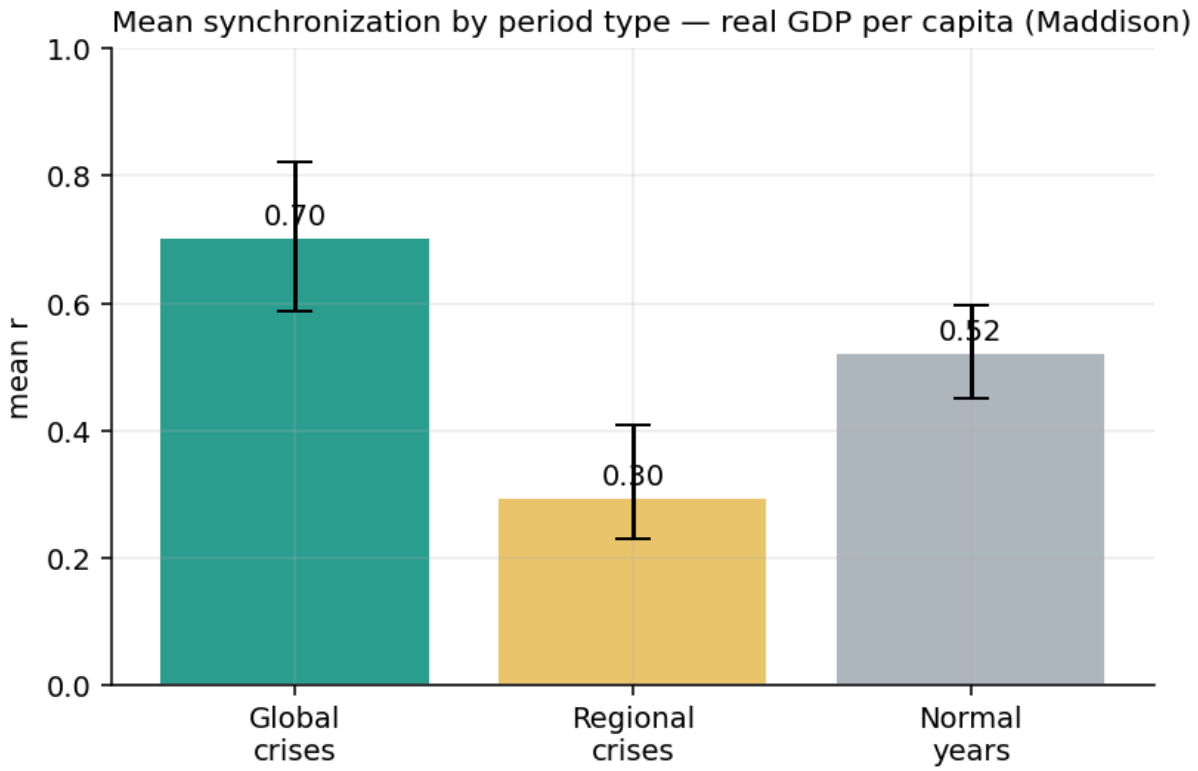


Fig 16. Mean synchronization by period type, real GDP. **Implication:** the gaps widen — global (0.70) > normal (0.52) >> regional (0.30) — confirming and sharpening the finding.

5 · External Dating I — Banking Crises

To remove any circularity from choosing crises by hand, crises are dated by an independent source (Jordà–Schularick–Taylor banking chronology, aligned with Laeven–Valencia).

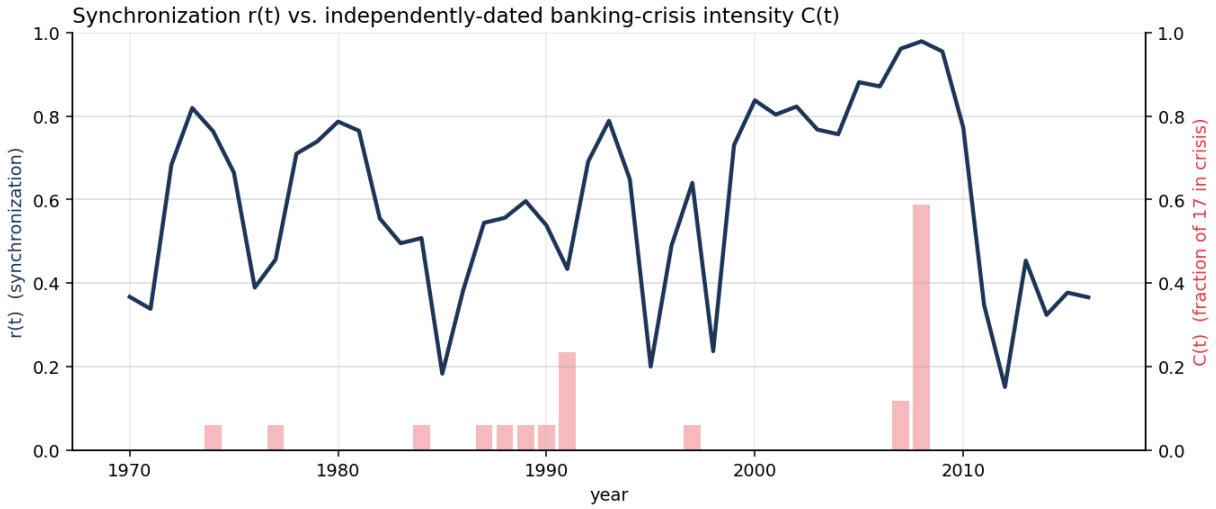


Fig 17. Synchronization vs. banking-crisis intensity C(t). Crises dated from bank failures, not GDP. **Implication:** this breaks the circularity of hand-picked windows — but the banking crises cluster on one global event (2008).

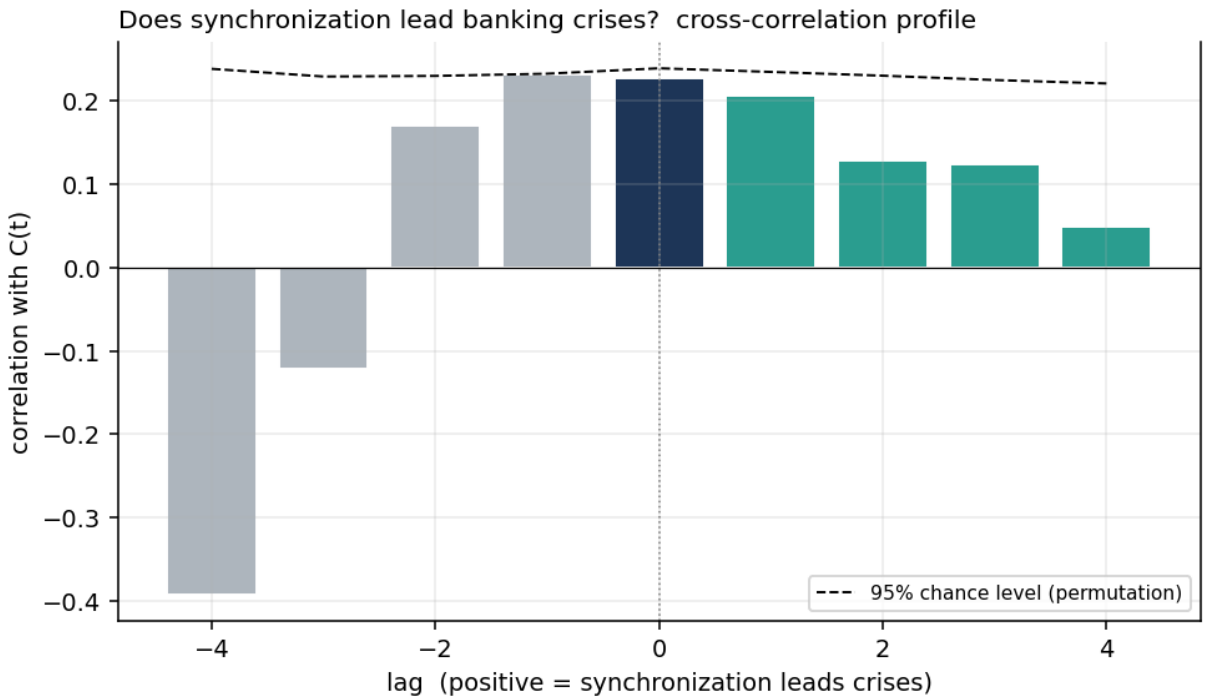


Fig 18. Lead-lag profile, banking crises. **Implication:** the correlation is modest and shows no clear lead — because there is essentially one global banking crisis, leaving little statistical power.

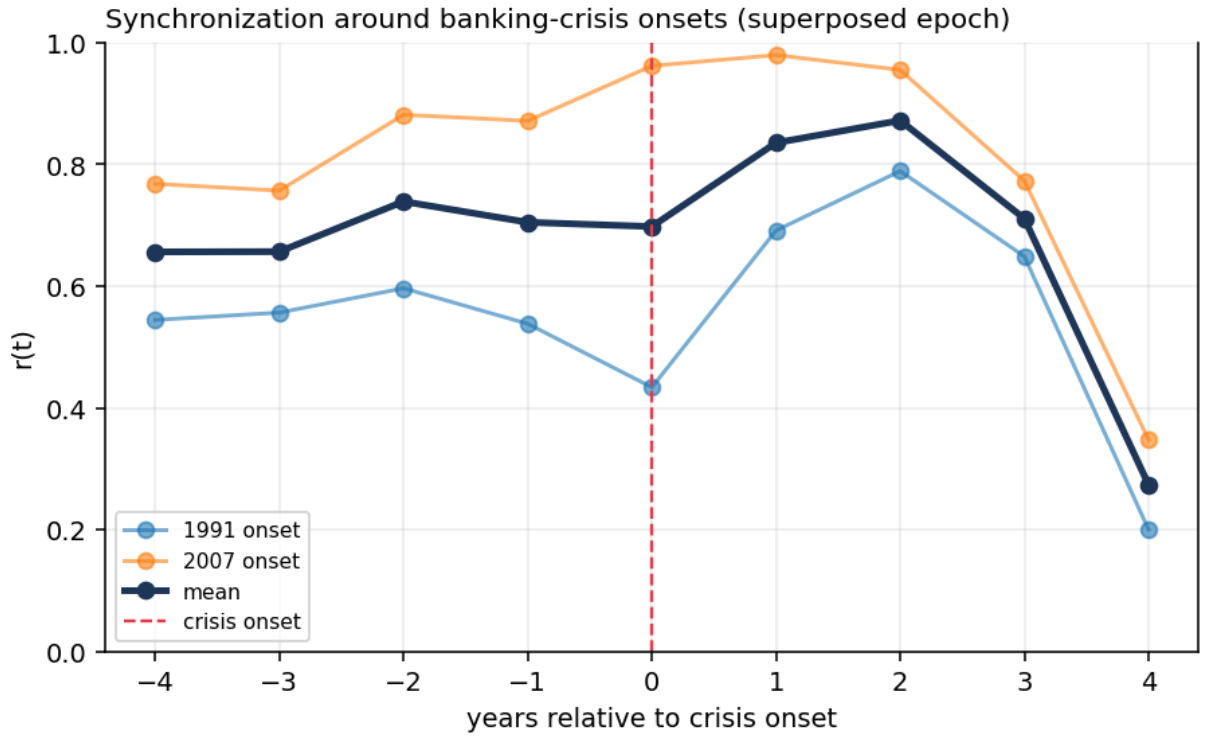


Fig 19. Synchronization around banking-crisis onsets. **Implication:** elevated at the 2007 onset but noisy across the mostly-regional banking crises.

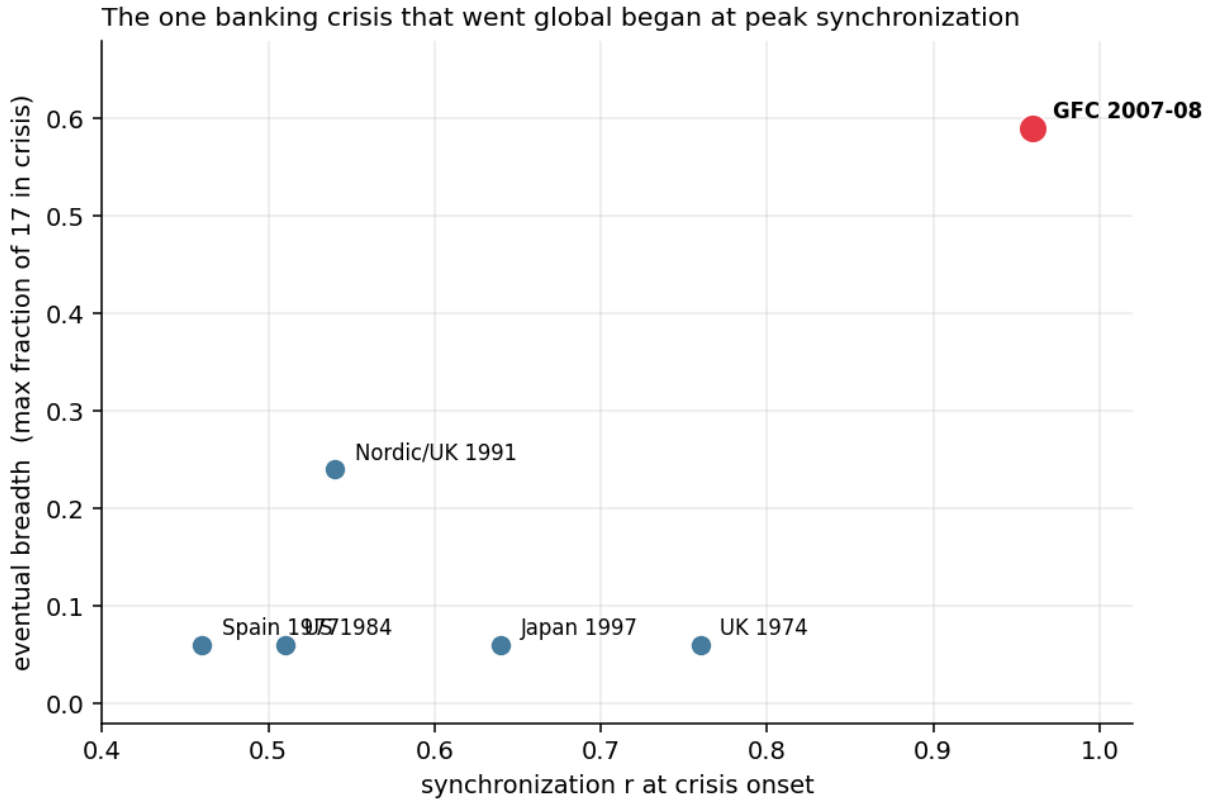


Fig 20. Onset synchronization vs. how global each banking crisis became. **Implication:** the one crisis that went global (2008) began at near-peak synchronization; every crisis that began at ordinary synchronization stayed contained — the loaded-gun idea, resting on a single event.

6 · External Dating II — Global Recessions (Primary Test)

The definition that fits fibrillation: Kose–Terrones global recessions — real-activity, externally dated, and five events (1975, 1982, 1991, 2009, 2020) rather than one. This is the paper's primary crisis test.

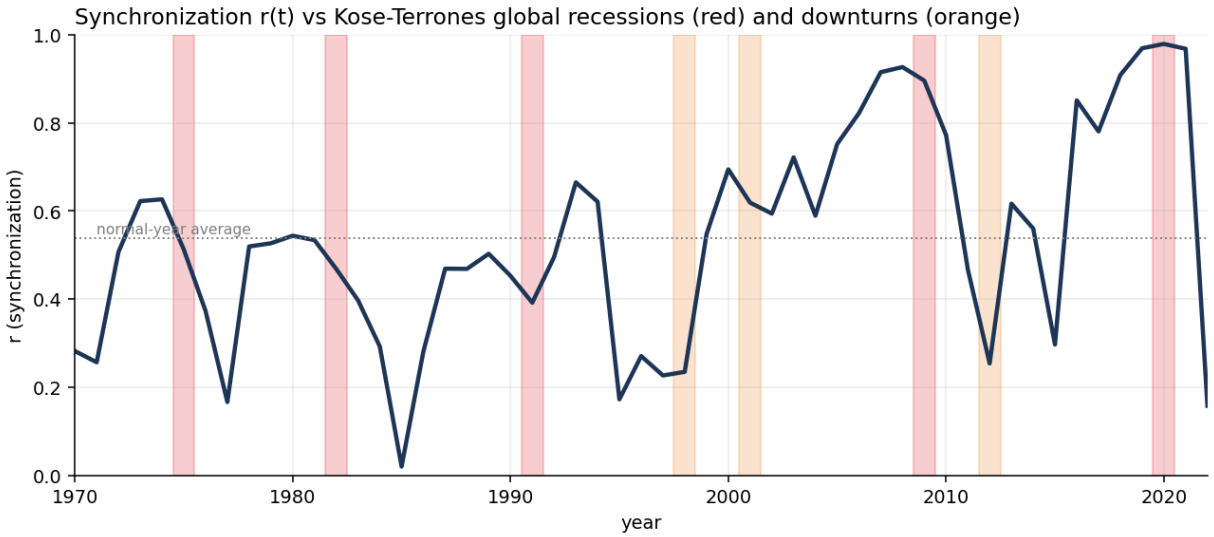


Fig 21. Synchronization vs. Kose–Terrones global recessions (red) and downturns (orange). **Implication:** five externally-dated, real-activity global crises restore the statistical power the banking definition lacked.

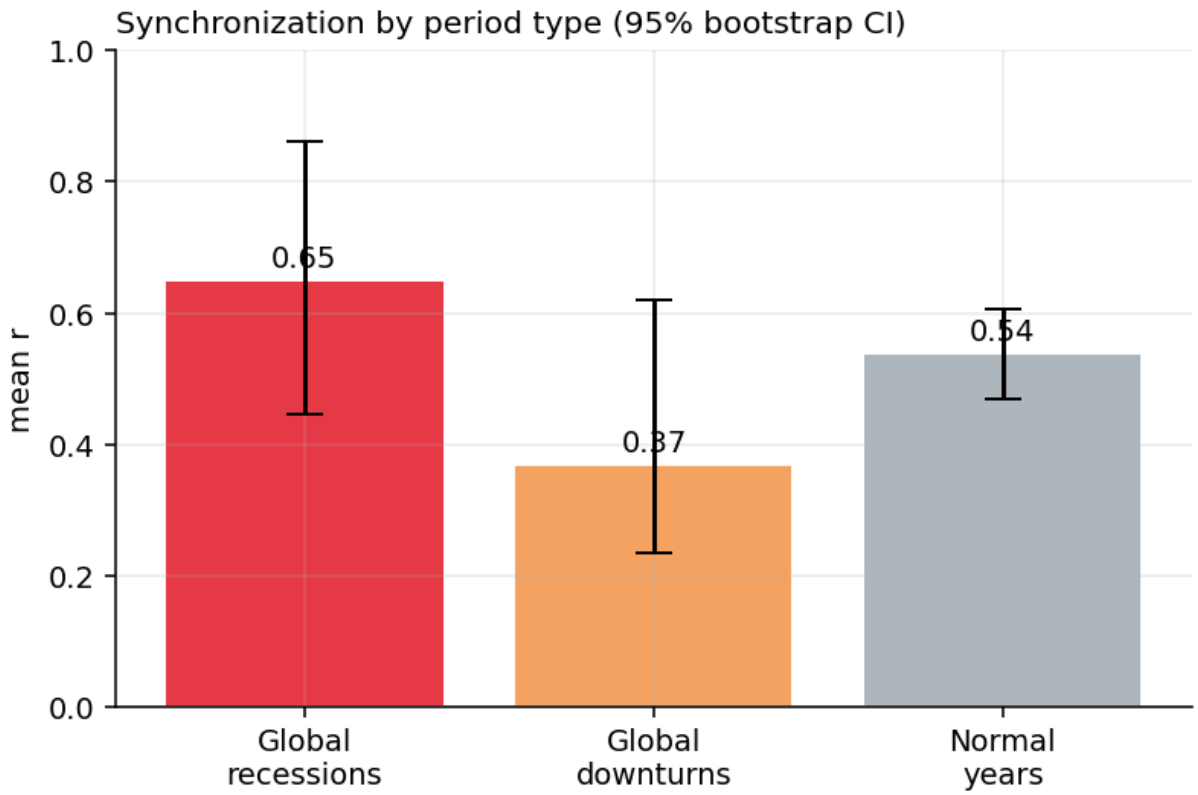


Fig 22. Mean synchronization by period type (global recessions). **Implication:** global recessions are the most synchronized (0.65); milder downturns the least (0.37), below normal years — synchronization tells true global recessions apart from ordinary slowdowns.

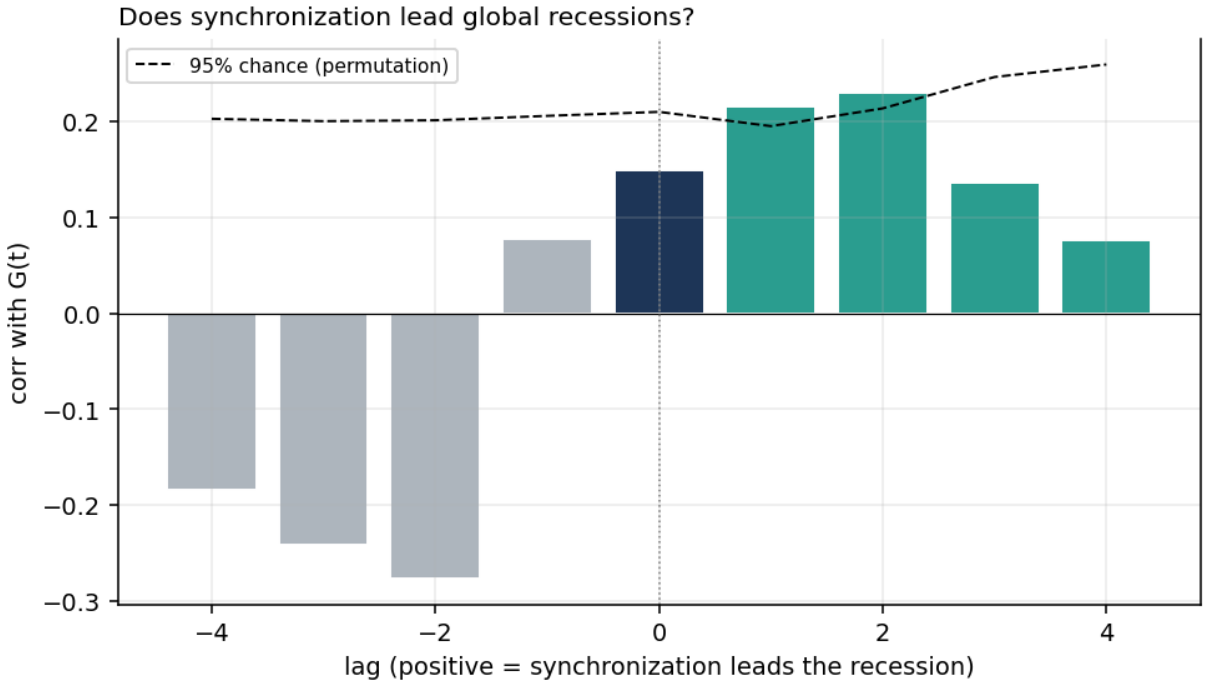


Fig 23. Lead-lag profile, global recessions. **Implication:** the correlation peaks at positive lags — synchronization leads the recession by one to two years, the signature of a genuine early warning.

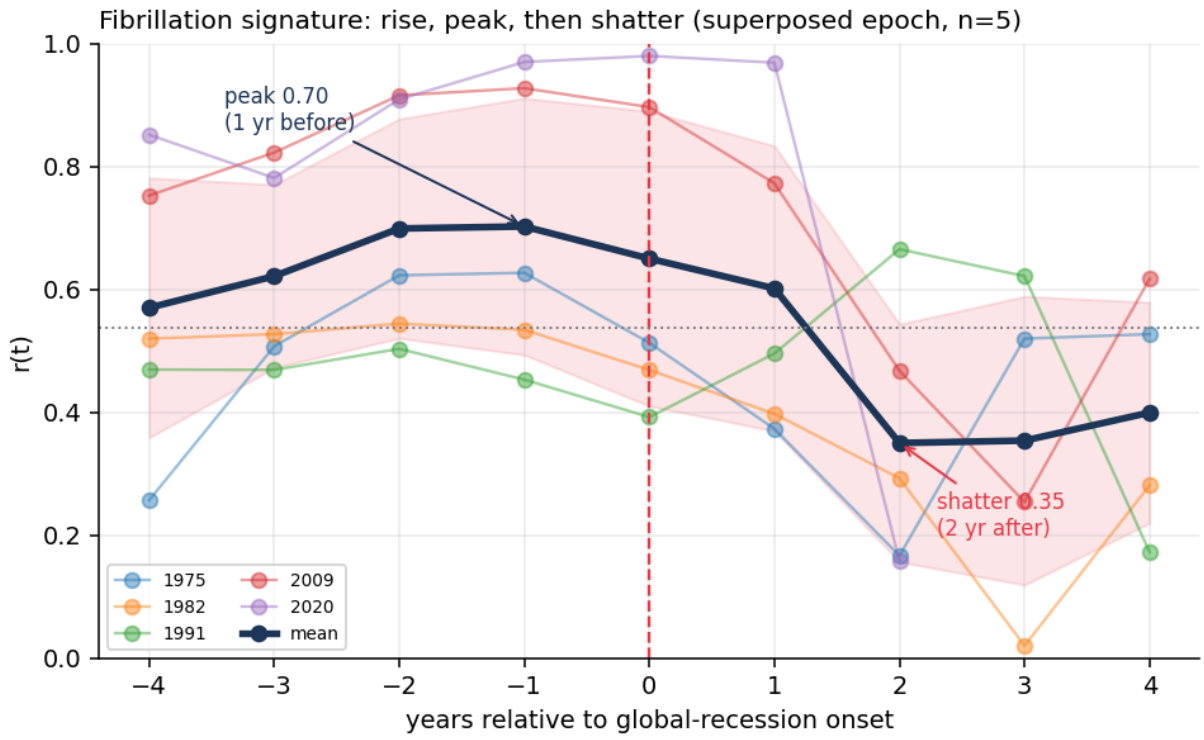


Fig 24. The fibrillation signature (superposed epoch, n=5). Mean synchronization around the five recession onsets, annotated with the pre-crisis peak and post-crisis shatter. **Implication:** synchronization climbs to a peak (0.70) the year before onset and collapses (0.35) two years after — build-up, ignition, rupture, on independent dates.

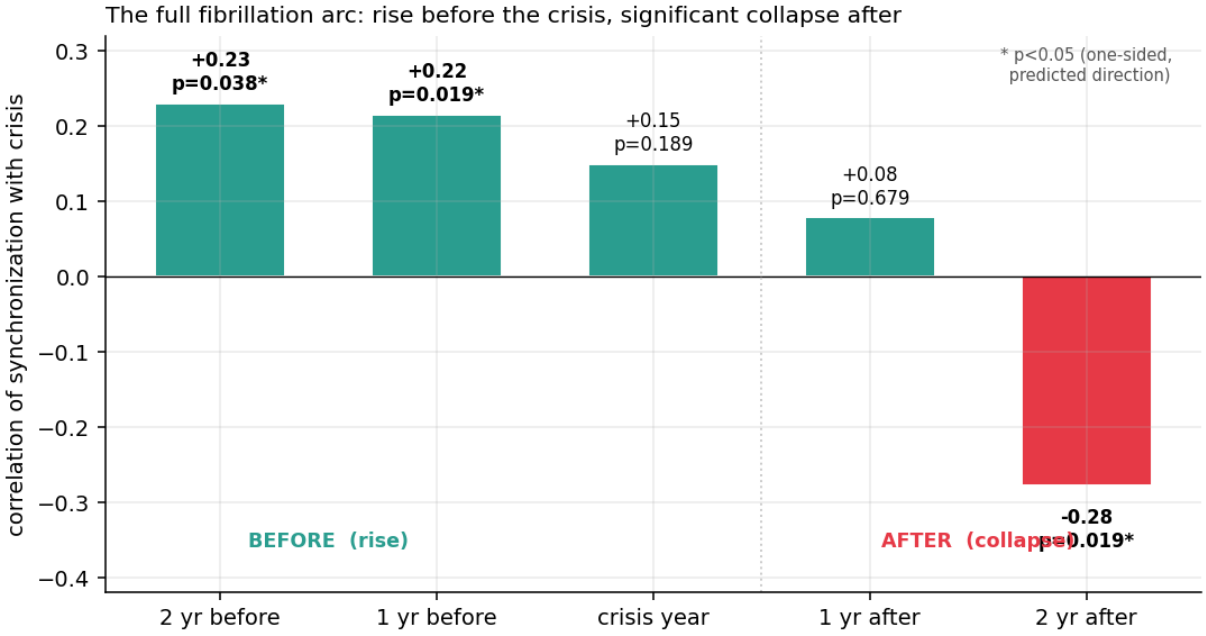


Fig 25. The full tested arc (two years before → two years after). Correlation of synchronization with the crisis at each timing. **Implication:** synchronization rises significantly before (p = 0.019, 0.038) and collapses significantly after (p = 0.019). The signature is statistically supported on both sides, and the lead/lag ends cannot be a circular artefact.

7 · Head-to-Head — Why the Real-Activity Definition

A direct comparison of the two crisis definitions, showing why the paper builds on global recessions rather than banking crises.

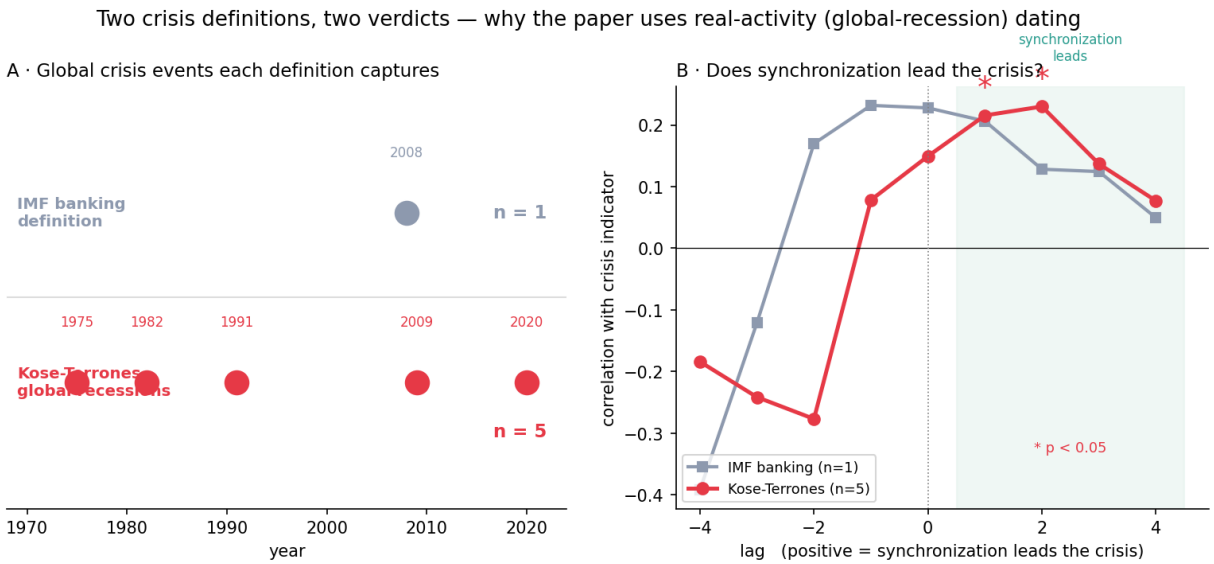


Fig 26. Two crisis definitions, two verdicts. Left: the banking definition captures one global event; the global-recession definition captures five. Right: their lead-lag profiles — only the real-activity definition shows a significant lead. **Implication:** fibrillation is a phenomenon of the real economy and must be dated by real activity.

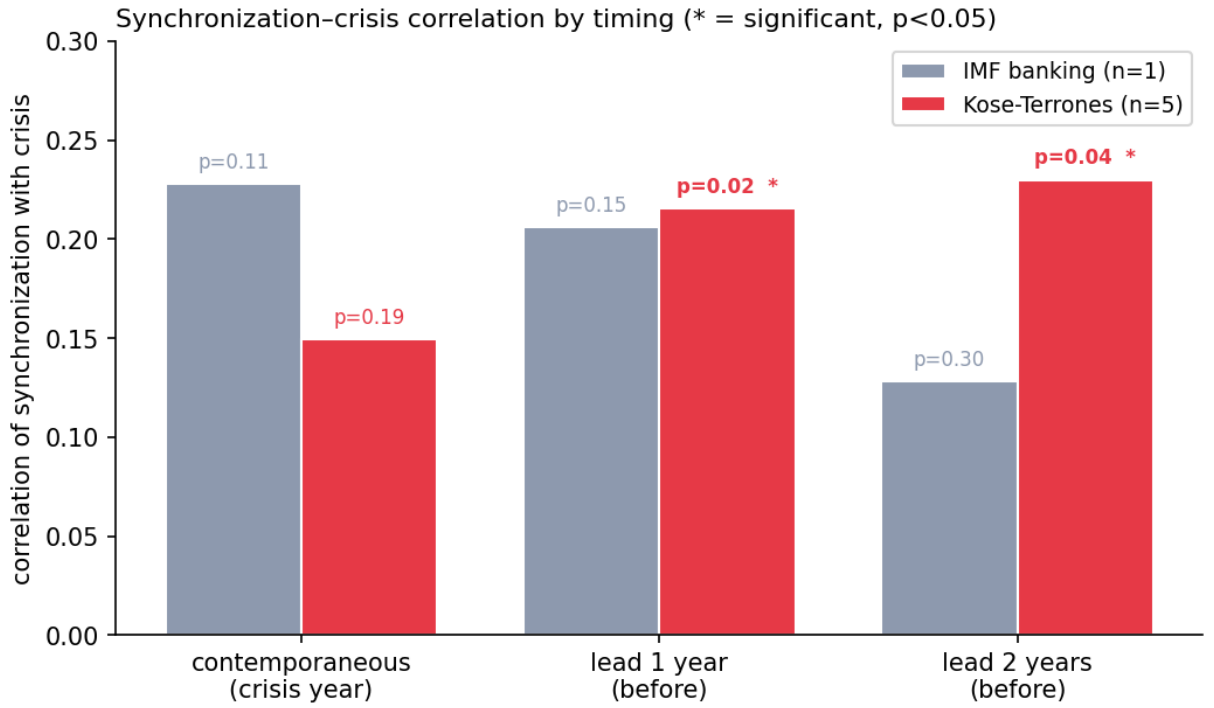


Fig 27. Synchronization–crisis correlation by timing, both definitions. **Implication:** the banking definition never reaches significance; the global-recession definition does, precisely at the leads — the clearest argument for the paper's chosen crisis definition.